

**Parietal vs. Frontal Lobe Cortical Gray Matter Volume
as Indicators of Alzheimer's Disease**

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1 Introduction

Alzheimer's is a neurodegenerative disorder that affects millions of people worldwide, and is characterized by loss of memory, difficulty performing simple cognitive functions, and the gradual deterioration of brain structure that significantly hinders one's ability to normally go about their daily life (NACC, 2025). This study aims to explore potential early indicators for Alzheimer's Disease utilizing biological features as well as elements related to an individual's characteristics; this study will primarily use the data and information gathered by the NACC as well as other sources that utilized their data.

1.1 Objective

Investigate the differences between the left parietal lobe cortical gray matter volume and the total frontal lobe cortical gray matter volume in the diagnosis of Alzheimer's while controlling for education.

1.2 Data Description

The data used in this analysis is a subset gathered from the National Alzheimer's Coordinating Center (NACC) database. This database comprises data from over 54,000 participants collected across more than 40 Alzheimer's Disease Research Centers (ADRCs) in America over the last 25+ years. The working sample consists of only 2,700 observations (participants) with 57 different variables. The variables that this analysis is concerned with are as follows:

Feature Name	Definition	Data Type	Possible Values
NACC ID	Unique identifier for each subject	N/A	N/A
DIAGNOSIS	Clinician diagnosis	Categorical	0 = Normal cognition 1 = Mild cognitive impairment 2 = Dementia due to AD
EDUC	Subject's number of years of education	Numeric	0 - 36
FRCORT	Segmented total frontal lobe cortical gray matter volume (cc)	Numeric	Regional gray matter volumes (94 - 278)
LPARCORT	Segmented left parietal lobe cortical gray matter volume (cc)	Numeric	Regional gray matter volumes (3 - 67)

*1 cc = 1 mL

2 Exploratory Data Analysis

This section provides an overview of the key variables in the analysis using numerical summaries and visualizations. Each variable is briefly described, followed by a preliminary look at the relationships between them.

2.1 Analyzing Individual Variables

In this section, each individual variable is described and given a preliminary analysis including overall distribution and observations. What is important to note is that the diagnosis variable was changed to 'diagnose' in which the only difference is that 1 and 2 diagnoses are classified as both being dementia due to Alzheimer's Disease.

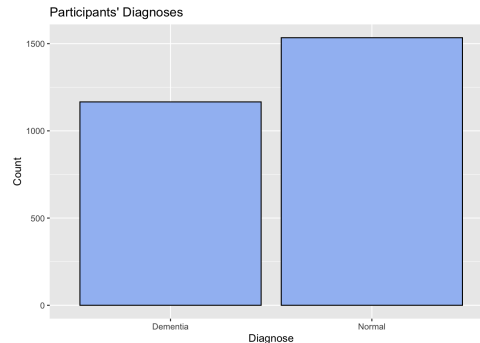


Figure 1.a
Counts of normal and dementia diagnoses

As shown in Figure 1.a the majority of participants from the sample were diagnosed with having normal cognition and showed no signs of having Alzheimer's. Although there is still a significant number of participants who were diagnosed with showing some signs of Alzheimer's with 1166 individuals compared to the number of normal diagnoses at 1534 individuals.

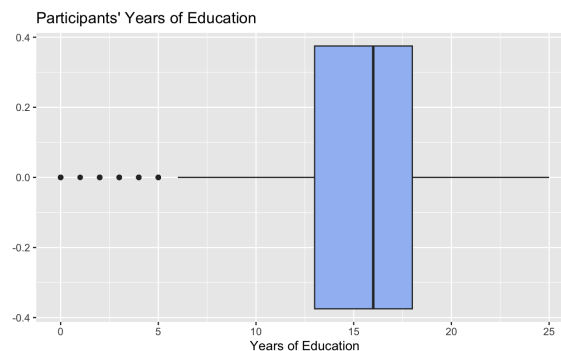


Figure 1.b
Boxplot of the participants' years of education

The large majority of participants had a post-secondary education with more than half of the sample having at least 12 years of education as shown in the Figure 1.b boxplot. The median education was 16 years with 50% of the participants having between 13 and 18, supporting the observation that most participants had at least some post-secondary education. Given the distribution, there were a few outliers with some participants having 5 or less years of education total; it is unclear which level of education these years account for.

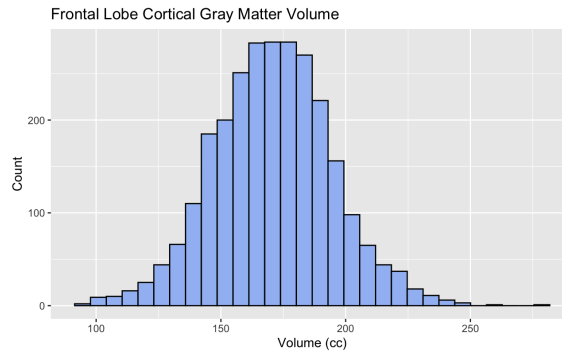


Figure 1.c
Histogram of frontal lobe cortical gray matter

The distribution of cortical gray matter volume in the frontal lobe is approximately normal as visualized in Figure 1.c, with minor outliers towards the right of the distribution. The mean sits at 171 cc, with a standard deviation of 23.7 cc, and the middle 50% of the sample's gray matter volume ranging between 155 cc and 186 cc.

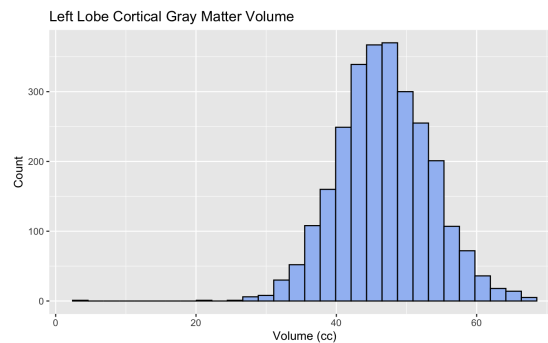


Figure 1.d
Histogram of left parietal lobe gray matter volume

The distribution in Figure 1.d of cortical gray matter volume in the left parietal lobe is also approximately normal, with minor outliers towards the left of the distribution. The mean falls around 46.7 cc. The standard deviation stood at around 6.53 cc and half of the sample of participants' cortical gray matter volume are primarily in the 42.5 - 51.2 cc range which are the 1st and 3rd quartiles.

2.2 Analyzing relationship between variables

In this section, the relationships between two or more variables were observed and analyzed through the use of side-by-side boxplots and scatterplots to visualize how cortical gray matter volume varies across diagnosis groups.

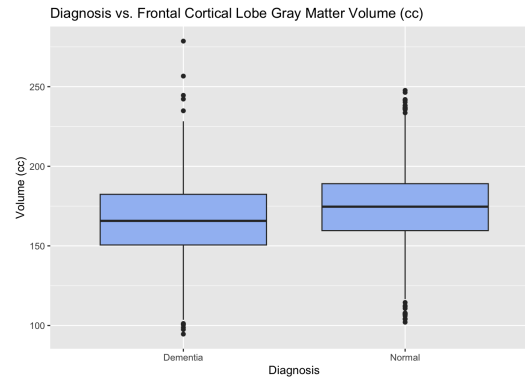


Figure 2.a

Boxplots of frontal lobe cortical gray matter volume within diagnosis groups

When looking at the relationship between the diagnoses and frontal lobe cortical gray matter volume, we can see in Figure 2.a there is a fairly symmetric distribution of gray matter volume of those who showed normal cognition, while the distribution for those diagnosed with dementia was somewhat more skewed. The median for frontal lobe cortical gray matter volume of a participant diagnosed with dementia was 166 cc, while those with normal cognition had a median of 175 cc of frontal lobe cortical gray matter volume. The IQRs were also similar (31.8 cc and 29.5 cc respectively), indicating comparable variability between groups.

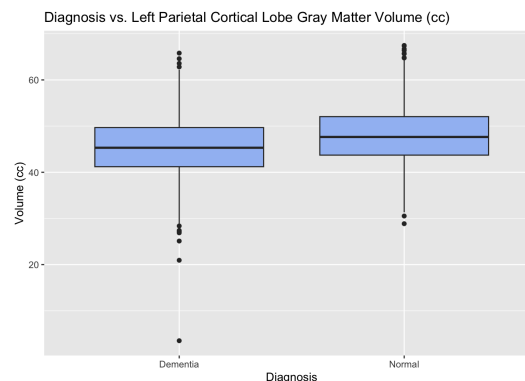


Figure 2.b

Boxplots of left parietal lobe cortical gray matter volume within diagnosis groups

With the left parietal lobe cortical gray matter volume in relation to diagnosis, the story is very similar as its frontal lobe counterpart indicated in Figure 2.b. Normal cognition diagnoses had a higher median gray matter volume at 47.6 cc, compared to the lower median of cortical gray matter volume in the left parietal lobe of those diagnosed with dementia at 45.3 cc. The IQRs of both groups were similar with normal cognition diagnoses being 8.3 cc and signs of dementia diagnoses being 8.4 cc.

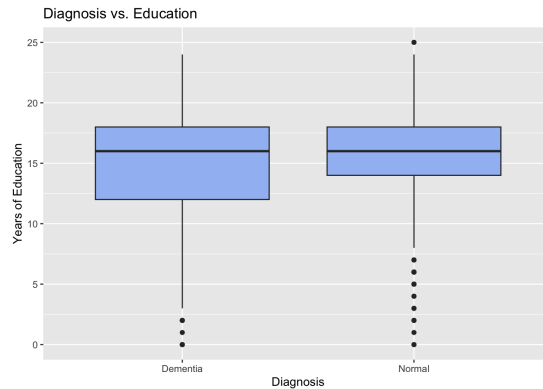


Figure 2.c
Boxplots of years of education within diagnosis groups

When it comes to education and its relationship with diagnosis both groups had identical median education levels at 16 years in Figure 2.c, indicating that years of schooling did not differ meaningfully in central tendency. However, the wider IQR of 6 years compared to 4 years in the normal cognition group suggests greater variability in education levels amongst those diagnosed with dementia. Both groups had low-end outliers, with several participants in each group reporting fewer than 5 years of education.

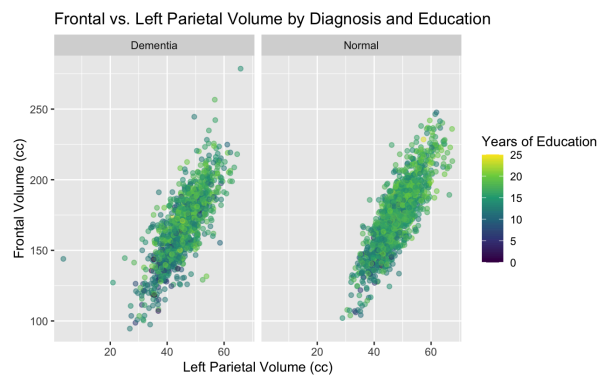


Figure 2.d
Scatterplots of frontal vs. parietal lobe gray matter volume by diagnosis and education

Figure 2.d shows a strong positive linear relationship between left parietal and frontal lobe cortical gray matter volume in both diagnosis groups. The Dementia group clusters slightly lower and to the left, while the Normal group extends further toward the upper right. Education shows no clear visual gradient within either panel, suggesting it does not visibly modify any parietal-frontal volume relationship.

3 Methods

This section outlines the statistical techniques used to analyze the relationships between cortical gray matter volume, education, and Alzheimer's diagnosis. Multiple linear regression and logistic regression are used to model the joint effect of frontal and left parietal volume on diagnosis while controlling for education. Additionally, inferential techniques include a hypothesis test for difference of means and a chi-square test of independence to assess group-level differences.

3.1 Regression

In this section, a multiple linear regression and logistic regression model are applied to examine the relationship between cortical gray matter volume, education, and Alzheimer's diagnosis. These models help provide a more complete picture of how our variables interact and how they contribute to predicting Alzheimer's diagnosis, beyond what is simply observed in the existing data.

3.1.1 Multiple Linear Regression

Multiple linear regression is used when the response variable that is being predicted is continuous. Similar to Figure 2.d, this analysis will utilize the primary variables of interest of the current study, with frontal lobe cortical gray matter being what is predicted (y) as a function of left parietal volume, education, and diagnosis serving as the predictor variables. The diagnose variable was recoded as logical (True = Dementia, False = Normal) for this analysis. The general multiple linear regression equation that the model will be derived from is as follows:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \epsilon_i$$

The respective predictor coefficients are represented as β for k explanatory variables used.

Regression Model Equation

Predicted frontal volume = 27.11 + 2.86(left parietal volume) + 0.65(education) – 0.37(diagnose)

Using this model from the `lm()` function in R, predictions can be made for an individual's frontal lobe cortical gray matter based on the three predictor variables. The strongest predictor for this model was the left parietal volume ($b_1 = 2.86$), suggesting that most of the explained variation in frontal volume stems from the left parietal and frontal volume relationship. For example, an individual who has 50 cc of left parietal lobe gray matter, 14 years of education, and was not diagnosed with dementia would be predicted to have a frontal lobe gray matter volume of 179.21 cc which is slightly above the sample average of 171 cc. Compared to the individual's actual frontal lobe volume of 182.81 cc, the resulting residual is 3.6 meaning the model underestimated the actual value.

For an individual with the same exact left parietal volume and education except as diagnosed with dementia, the model predicts they would have a frontal lobe gray matter volume of 178.74 cc. Compare this to the first prediction made by the equation and there is not much of a noticeable difference between the two other than dementia diagnosed individuals are likely to have lower frontal lobe gray matter which is consistent with the earlier observations made in the exploratory data analysis comparing the diagnosis groups and their gray matter volumes.

Model Evaluation

The adjusted R^2 for the model was 0.674, meaning that with the specific explanatory variables used, 67.4% of the variation was accounted for by the model given the many unmeasured factors. The Root Mean Squared Error (RMSE) for the model was 13.53 cc meaning that the predictions deviated from actual values by approximately 13.53 cc on average. Overall, the model is moderately strong and reliable in predicting frontal lobe gray matter volume.

3.1.2 Logistic Regression

Logistic regression is used when the response variable that is being predicted is binary, as in the case of this study. This analysis will address the main objective of this study in determining the strength of gray matter volume in the left parietal and frontal lobes alongside education as predictors for Alzheimer's Disease diagnosis. Because the response is bounded between 0 and 1 (a probability), the logit link function is used to model the log-odds of dementia diagnosis as a linear combination of the predictors; the resulting coefficients are on a log-odds scale meaning they change the odds of success by a factor of $e^{(\text{coefficient})}$. The general logit form equation is as follows:

$$\log(p_i / (1 - p_i)) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki}$$

Because the primary objective is determining the likelihood of a diagnosis rather than the logit, this study opted to use the probability form of the logit form equation which is more interpretable in the current context.

$$p_i = \frac{e^{\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki}}}{1 + e^{\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki}}}$$

Similar to the multiple regression model equation, the respective predictor coefficients are represented as β for as many k explanatory variables are used to generate a probability (p) for being diagnosed with dementia.

Regression Model Equation

$$\log(p_i / (1 - p_i)) = 3.08 - 0.002(\text{frontal volume}) - 0.05(\text{left parietal volume}) - 0.04(\text{education})$$

Using the `glm()` function in R, the model estimates the probability of an individual being diagnosed with dementia based on the three predictor variables. All three predictors have negative coefficients, meaning higher values of brain volume and education are associated with lower odds of dementia which supports the broader hypothesis that greater cortical gray matter volume and cognitive reserve may help prevent Alzheimer's-related diseases. Using the model with an example, an individual with 180 cc of frontal volume, 50 cc of left parietal volume, and 14 years of education would have predicted log-odds of -0.34. Converting to a probability gives $p \sim 41\%$, indicating the model predicts a roughly 41% chance of dementia diagnosis for this individual, which points in the right direction as this individual was diagnosed with normal cognition.

Using an individual with slightly higher frontal and left parietal gray matter volumes at 200 and 60 cc respectively while keeping education at 14 years, the probability generated by the model comes out to be $p \sim 29\%$. This difference within education levels shows that regardless of education level, it is generally less likely to have dementia based more on gray matter volumes which lines up with the observations in our multiple linear regression model.

3.2 Inferential Analysis

This section will explain and apply inferential techniques to test the patterns observed in the exploratory analysis. These techniques include a two-sample hypothesis test for the difference of means between the two diagnosis groups and a chi-square test of independence for the categorical data; the Central Limit Theorem ($n > 30$) is met, and independence and random sampling are assumed for both tests.

3.2.1 Hypothesis tests for difference of means

Two sample t-tests for the difference of means are used to determine whether or not two samples' averages of a particular variable are significantly different. In this analysis, the volume of cortical gray matter in both the frontal and left parietal lobe is compared between the two diagnosis groups. To reject the null hypothesis for difference in means, the p-value needs to be below the significance level and the confidence interval (CI) must not include 0.

Frontal Lobe Cortical Gray Matter

Hypotheses

$$H_0: \mu_{AD} = \mu_{Normal}$$

$$H_A: \mu_{AD} < \mu_{Normal}$$

Summarized Data

$$\bar{x}_{AD} = 166.25 \text{ cc}, s_{AD} = 23.74 \text{ cc}$$

$$\bar{x}_{Normal} = 174.48 \text{ cc}, s_{Normal} = 23.08 \text{ cc}$$

Left Parietal Lobe Cortical Gray Matter

Hypotheses

$$H_0: \mu_{AD} = \mu_{Normal}$$

$$H_A: \mu_{AD} < \mu_{Normal}$$

Summarized Data

$$\bar{x}_{AD} = 45.37 \text{ cc}, s_{AD} = 6.56 \text{ cc}$$

$$\bar{x}_{Normal} = 47.93 \text{ cc}, s_{Normal} = 6.29 \text{ cc}$$

Results and Conclusion

Using the `t.test()` function in R, Welch's t-tests were conducted for both regions, with cortical gray matter volume as the numerical variable and diagnosis as the categorical variable. For the frontal lobe, $t(2470.7) = -9.033$, $p < .001$, 95% CI [-10.020, -6.445]. For the left parietal lobe, $t(2453.6) = -10.212$, $p < .001$, 95% CI [-3.049, -2.067]. In both cases, the p-value is below $\alpha = 0.05$ and the confidence interval excludes 0, so the null hypothesis is rejected for each region. The data provide sufficient evidence that mean cortical gray matter volume in both the frontal and left parietal lobes is significantly lower in participants diagnosed with dementia than in those with normal cognition, supporting the broader analysis objective that gray matter volume in these regions serves as a meaningful indicator of Alzheimer's Disease.

3.2.2 Chi-square Test of Independence

A chi-square test of independence is used to determine if two (or more) categorical variables are independent or significantly associated; in the case of the current study, education level's association with Alzheimer's diagnosis is tested. For this test, education was transformed from a numerical variable to a categorical variable. The assumed specific education level groups are as follows:

Years of education	< 12 years	$12 \leq \text{years} < 16$	≥ 16
Level	Less than High school	High school to some college	Bachelor's or Higher

Figure 3.a
Coding of education levels

Summarized Data

	Dementia	Normal	Sum
Bachelor's or Higher	624	961	1538
Highschool to some college	435	504	939
Less than Highschool	107	69	176
Sum	1166	1534	2700

Figure 3.b
Contingency table of education level by diagnosis

Hypotheses

H₀: Education level and diagnosis are independent

H_A: Education level and diagnosis are not independent

Results and Conclusion

Using the `chisq.test()` function in R, a chi-square test of independence was conducted with education level (3 categories) and diagnosis as the two categorical variables. Running the test resulted in the following: $\chi^2(2) = 35.43, p < .001$. Because the p-value is less than the significance level of $\alpha = 0.05$, the null hypothesis is rejected, and the data provide sufficient evidence that education level and diagnosis are associated; this finding supports the objective in establishing education as a significant explanatory variable for diagnosis that should be controlled for to prevent confounding results. Based on these results and the data from Figure 3.c, individuals with lower education levels appear to be more frequently diagnosed with Alzheimer's in this sample.

4 Results

The findings of this analysis support the main objective of investigating and identifying left parietal lobe and total frontal lobe cortical gray matter volume as meaningful indicators in the diagnosis of Alzheimer's while also observing education as a third contributing variable. As observed in the exploratory data analysis, more gray matter volume was generally associated with reduced odds of having dementia; contrarily, an individual was more likely to have dementia with lower volumes of gray matter in both brain areas. Using the t-tests, this confirmed the case as participants diagnosed with dementia had significantly lower median frontal and left parietal cortical gray matter volumes compared to those with normal cognition. With this finding, both frontal and left parietal cortical gray matter volume serve as significant indicators of Alzheimer's Disease.

Education did not strongly differ between diagnosis groups in central tendency, however, conducting a chi-square test of independence revealed a statistically significant association when

	Dementia	Normal
Bachelor's or Higher	.393	.606
Highschool to some college	.463	.537
Less than Highschool	.608	.392

Figure 3.c
Row proportions of diagnosis within each education level

classifying years of education into levels. The proportion of dementia diagnoses was notably higher among participants with less than a high school education compared to those with some post-secondary education, consistent with the observations of cognitive reserve theory in which regular mental stimulation may help improve and maintain cognitive functioning especially in older individuals (Stern, 2012). Even so, education's effect appears to be more indirect than that of cortical gray matter volume suggesting it may play a less direct role in distinguishing an individual's cognitive status in this sample.

5 Future Work

For future research, addressing limitations such as a limited dataset is recommended to support more in-depth analysis similar to the current study; this was especially relevant given the assumption of education level having to be established for the chi-square test. Furthermore, reporting effect size alongside the p-values would help strengthen interpretation in future work. For example, using Cohen's d as a measure for effect size would provide a practical magnitude of effect for the observed statistical results from the t-tests; a Cramér's V could also be applied for the chi-square test results. Additionally, utilizing the most accurate data for future research would be an improvement as modern medicine for Alzheimer's is constantly advancing; sourcing up-to-date data and information is recommended to most accurately capture the external validity.

6 References

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Stern, Y. (2012). Cognitive Reserve in Ageing and Alzheimer's Disease. *The Lancet Neurology*, 11(11), 1006–1012. [https://doi.org/10.1016/S1474-4422\(12\)70191-6](https://doi.org/10.1016/S1474-4422(12)70191-6)